Break-over diode (BOD): device, application circuits and Simscape model `dtt.crowbar.bod`

Rev. 00 – 2026-09-22

D. Bagnara

2026-09-22

Table 1: Revision history

Rev.	Date	Description
00	2026-09-22	First issue: device description, application circuits, functional Simscape model with switching dynamics and optional thermal port, verification with a Python replica of the model equations. Default data IXBOD2-11.

Contents

1 Scope **2**

2 The break-over diode **2**
2.1 Structure and operation . . . 2
2.2 Static characteristic . . . 2
2.3 Dynamic characteristics . . . 3
2.4 Datasheet data of the default device . . . 4

3 Application: overvoltage protection and self-firing of thyristors **4**
3.1 Principle . . . 4
3.2 Design rules . . . 5

4 The Simscape model `dtt.crowbar.bod` **6**
4.1 Structure and modes . . . 6
4.2 Equations . . . 7
4.3 Parameters . . . 9
4.4 Implementation notes . . . 9
4.5 Limitations . . . 10

5 Verification with the Python replica **10**

6 Use in the crowbar models **11**

A Listing of `bod.ssc` **12**

1 Scope

This note describes the break-over diode (BOD), the two-terminal p-n-p-n device used as a passive overvoltage trigger for thyristors, and documents the Simscape component `dtt.crowbar.bod` written for the DTT DIV crowbar models. The component is a functional model (piecewise-linear characteristics, event-based mode switching) with the dynamics that matter for a trigger device: turn-on time, $\mathrm{d}v/\mathrm{d}t$ triggering, holding-current turn-off and a surrogate of the turn-off time t_q. Temperature dependence of the breakover voltage, holding current, off-state current and critical $\mathrm{d}v/\mathrm{d}t$ is included, with an optional thermal port.

The model equations were verified with a Python replica of the component (`bod_sim.py`), which also produced the figures of Section 5. The `.ssc` file itself has not yet been compiled in MATLAB and must be checked with `ssc_build`. The default parameters are those of the Littelfuse/IXYS IXBOD2-11 ($1100\,\mathrm{V} \pm 50\,\mathrm{V}$), the device foreseen as backup trigger in the DTT crowbar. Parameters not given in the datasheet are marked as assumed.

2 The break-over diode

2.1 Structure and operation

A BOD is a four-layer p-n-p-n silicon device with two terminals, anode (A) and cathode (K), i.e. a thyristor without gate (a Shockley diode) engineered so that its forward breakover voltage V_{BO} is accurate, stable and reproducible. Figure 1 shows the symbol, the layer sequence and the two-transistor equivalent. In forward blocking (anode positive) the outer junctions J1 and J3 are forward biased and the central junction J2 blocks. The leakage current of J2, multiplied by the avalanche factor $M(v)$ of that junction, feeds the bases of the p-n-p and n-p-n transistors; the device switches to the on-state when the loop gain reaches unity,

$$M(v)\,\big[\alpha_1(i) + \alpha_2(i)\big] = 1. \tag{1}$$

In a thyristor this condition is normally reached by injecting gate current (which raises α_2); in a BOD it is reached by the avalanche multiplication $M(v)$ of J2 alone. The breakover voltage is therefore fixed by the avalanche breakdown of J2, which depends on doping and thickness of the n^- base and is well controlled by the process; commercial devices are specified with $\pm 50\,\mathrm{V}$ on V_{BO} and a positive temperature coefficient of the order of 0.7×10^{-3}–$2 \times 10^{-3}/\mathrm{K}$ [3, 4]. Cathode shorts and guard rings shape the current path so that breakover starts uniformly and the device can withstand the high initial $\mathrm{d}i/\mathrm{d}t$ of a trigger pulse [5].

Two properties distinguish a BOD from a thyristor. First, the small die (a trigger device: 0.9 A average, 200 A surge for 0.1 ms [3]) gives a fast turn-on: the anode voltage collapses within 0.3–0.5 µs of reaching V_{BO} [5]. Second, the reverse blocking capability is very low: the emitter junctions are heavily doped and shorted, so a discrete BOD element blocks only about 10 V in reverse [2]; whenever the application applies reverse voltage, a series diode is mandatory.

2.2 Static characteristic

The static i–v characteristic (Fig. 2) has the S-shape of all p-n-p-n devices: it is single-valued in current (the voltage is a function of the current) and multi-valued in voltage. Four regions are distinguished.

- *Forward blocking*, $0 < v < V_{\mathrm{BO}}$: the current is the off-state current I_D (0.2 µA at 25 °C, 200 µA at 125 °C, both at $0.8\,V_{\mathrm{BO}}$ for the IXBOD2), rising steeply near V_{BO} because of the avalanche multiplication. The breakover point is $(V_{\mathrm{BO}}, I_{\mathrm{BO}})$ with $I_{\mathrm{BO}} \le 20\,\mathrm{mA}$.
- *Negative resistance*, from $(V_{\mathrm{BO}}, I_{\mathrm{BO}})$ to the holding point $(V_{\mathrm{H}}, I_{\mathrm{H}})$: unstable for any source with an impedance lower than the magnitude of the negative slope; the operating point jumps along the load line to the on-state.

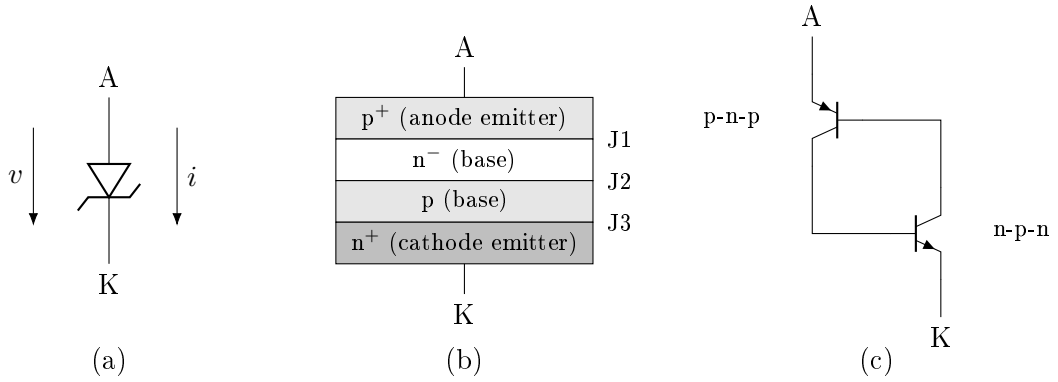


Figure 1: Break-over diode: (a) symbol used in this note (two-terminal breakover symbol), (b) four-layer structure, (c) two-transistor equivalent. Breakover occurs when the avalanche multiplication of J2 (the collector junction common to both transistors) makes the loop gain reach unity, Eq. (1).

- *On-state*, $i > I_{\mathrm{H}}$: $v = V_{\mathrm{T0}} + r_{\mathrm{T}}\, i$ with $V_{\mathrm{T0}} \approx 0.9\,\mathrm{V}$, $r_{\mathrm{T}} \approx 80\,\mathrm{m\Omega}$; $V_T = 1.3\,\mathrm{V}$ at $5\,\mathrm{A}$. The holding voltage V_{H} (4–8 V) is the voltage at I_{H}, before the full conductivity modulation.
- *Turn-off* occurs when the current falls below the holding current I_{H} (40 mA at 25 °C, 20 mA at 125 °C); the device then returns to the blocking branch.
- *Reverse*: blocking up to $V_{RB} \approx 10\,\mathrm{V}$ only, then breakdown of the emitter junctions.

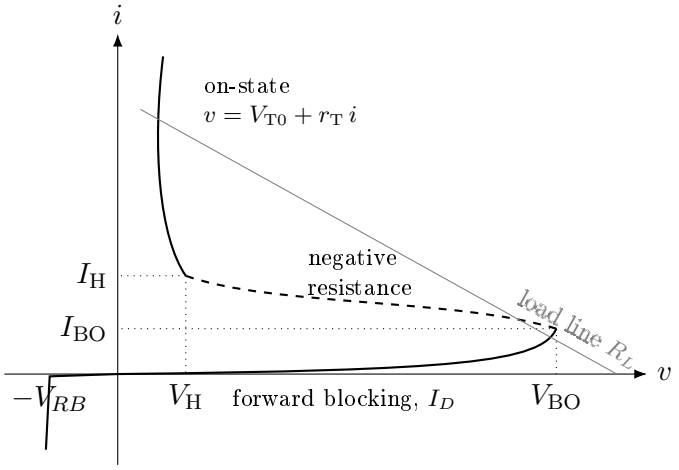


Figure 2: Qualitative static characteristic of a BOD. With a source of open-circuit voltage above V_{BO} and series resistance R_L the operating point jumps from $(V_{\mathrm{BO}}, I_{\mathrm{BO}})$ to the on-state along the load line; it jumps back to the blocking branch when the current falls below I_{H}.

2.3 Dynamic characteristics

Turn-on. Once v reaches V_{BO} the regenerative process of Eq. (1) takes a time of the order of the carrier transit time through the n^- base; the datasheets give no figure, the IXYS patent literature quotes a turn-on time (anode voltage collapse) of about 0.5 µs for conventional dice and below 0.3 µs for the thin epitaxial ones [5]. During this time the BOD behaves approximately as a voltage clamp at V_{BO}: the current is set by the external circuit while the voltage collapses.

$\mathrm{d}v/\mathrm{d}t$ triggering. A fast rise of the anode voltage drives the displacement current $C_j\,\mathrm{d}v/\mathrm{d}t$ of J2 through the bases of the two transistors; if this current, multiplied by $M(v)$, is large enough the device breaks over below V_{BO}. The datasheet gives a critical rate of $3000\,\mathrm{V/\mu s}$ at $v = 0.9\,V_{\mathrm{BO}}$, 25 °C

(1000 V/µs at 150 °C in older revisions) [3]. The capability is much lower at low voltage because $M(v)$ is small there; conversely, near V_{BO} a modest $\mathrm{d}v/\mathrm{d}t$ is enough. In thyristor protection circuits this is harmless (the BOD only fires the thyristor a little early), but it matters when the BOD stack is exposed to the switching $\mathrm{d}v/\mathrm{d}t$ of a converter, as happened with the DTT crowbar [8].

$\mathrm{d}i/\mathrm{d}t$ and surge. The trigger current must be limited by the series resistor: 150 A/µs repetitive, 500 A/µs non-repetitive, $I_{SM} = 200\,\mathrm{A}$ and $I^2t = 4\,\mathrm{A^2s}$ for 0.1 ms [3]. The junction temperature rise per pulse is estimated with the energy coefficient $K_P = 700\,\mathrm{K/J}$, i.e. an adiabatic thermal capacitance of 1.43 mJ/K.

Turn-off time. After conduction the stored charge must recombine before the full V_{BO} can be blocked again: $t_q = 200\,\mu\mathrm{s}$ (typical, 150 µs for the IXBOD1) [3, 4]. A forward voltage re-applied earlier re-triggers the device at a lower voltage.

2.4 Datasheet data of the default device

Table 2 collects the IXBOD2-11 data used for the default parameters [3], complemented with the IXBOD1 application data [2]. Where the two datasheet revisions found differ (e.g. I_{BO} 15 or 20 mA, K_T 0.7 or 1×10^{-3}/K) the values of the current Littelfuse revision were taken.

Table 2: IXBOD2-11 data (Littelfuse/IXYS) [3, 2]

Symbol	Quantity	Value	Conditions
V_{BO}	Breakover voltage	1100(50) V	$T_{\mathrm{j}} = 25\,^{\circ}\mathrm{C}$
K_T	Temperature coefficient of V_{BO}	1×10^{-3}/K	
I_{BO}	Breakover current	20 mA max	$T_{\mathrm{j}} = 25\,^{\circ}\mathrm{C}$
I_D	Off-state current	0.2 µA / 200 µA max	$0.8\,V_{\mathrm{BO}}$, 25 / 125 °C
I_{H}	Holding current	40 mA / 20 mA max	25 / 125 °C
V_{H}	Holding voltage	4–8 V	25 °C (IXBOD1 data)
V_T	On-state voltage	1.3 V max	$I_T = 5\,\mathrm{A}$, 25 °C
V_{T0}, r_{T}	Threshold voltage, slope resistance	0.9 V, 80 mΩ	150 °C
V_{RB}	Reverse blocking voltage	10 V max	IXBOD element [2]
$(\mathrm{d}v/\mathrm{d}t)_{cr}$	Critical rate of rise of off-state voltage	3000 V/µs	$0.9\,V_{\mathrm{BO}}$, 25 °C
$(\mathrm{d}i/\mathrm{d}t)_{cr}$	Critical rate of rise of on-state current	150 / 500 A/µs	repetitive / non-repetitive
t_q	Turn-off time	200 µs typ.	
$I_{F(AV)M}$, I_{RMS}	Average, RMS current	0.9 A, 1.4 A	50 Hz, $T_{amb} = 50\,^{\circ}\mathrm{C}$
I_{SM}, I^2t	Surge current, I^2t	200 A, $4\,\mathrm{A^2s}$	$t_p = 0.1\,\mathrm{ms}$
K_P	Energy coefficient	700 K/J	$\Delta T_{\mathrm{j}} = K_P\,E_{pulse}$
$R_{th(j\text{-}a)}$	Thermal resistance junction–ambient	60 / 45 K/W	natural / 2 m/s air
T_{j}	Junction temperature range	−40–150 °C	

3 Application: overvoltage protection and self-firing of thyristors

3.1 Principle

Figure 3 shows the classical BOD protection circuit [1, 2]. The BOD is connected between the anode and the gate of the thyristor T through the current-limiting resistor R_1 and the series diode D_1; the gate network $R_2 \parallel C_2$ and the gate lead to the gate unit are the usual ones. When the anode–cathode voltage reaches V_{BO} (plus the small drops on D_1 and on the gate), the BOD breaks over and a gate current $i_G \approx (v_{AK} - V_T)/R_1$ flows: the thyristor turns on after its gate-controlled delay time t_{gd}, the anode voltage collapses and the BOD current falls below I_{H}, so that the BOD turns off by itself. The thyristor is thus fired *before* its own breakover voltage is reached, in a controlled way through the gate rather than through the uncontrolled bulk breakover, which would concentrate the current in a small area and destroy the device. The same circuit serves as the passive self-firing network of a crowbar: the BOD sets the voltage at which the crowbar closes without any electronics.

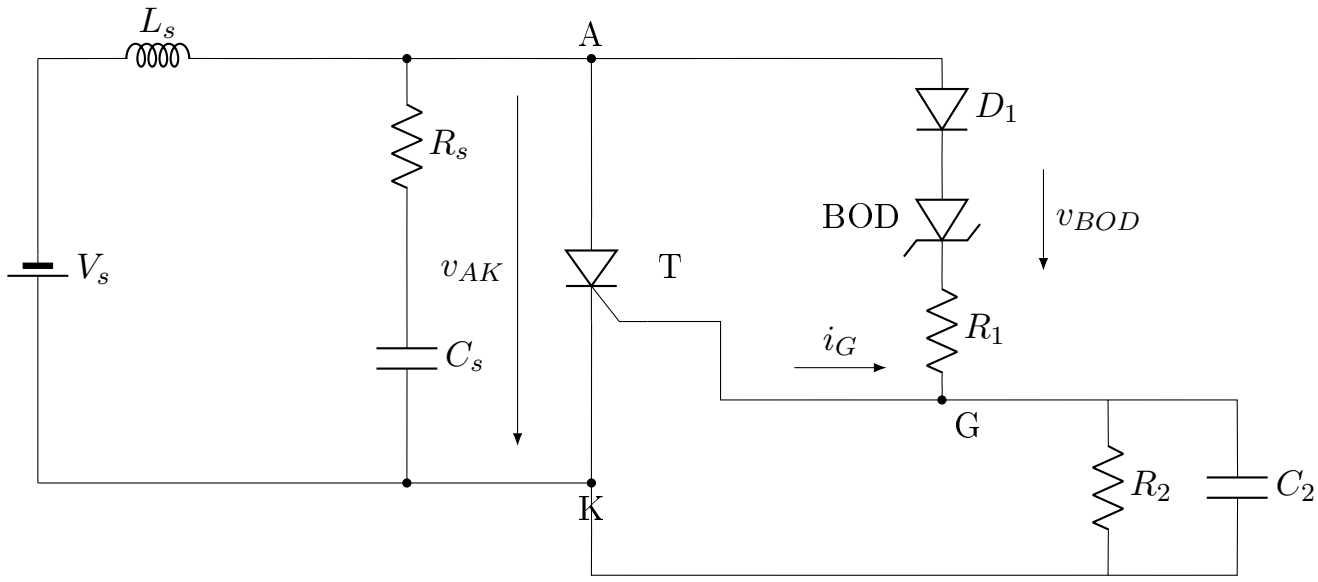


Figure 3: BOD overvoltage protection / self-firing of a thyristor: series diode D_1 (takes the reverse voltage, the BOD blocks only about 10 V), BOD, current-limiting resistor R_1 (0–200 Ω), gate network $R_2 \parallel C_2$ (100–1000 Ω, 22–47 nF) against spurious triggering [2]. R_s–C_s is the usual thyristor snubber.

The sequence of events after the anode voltage exceeds V_{BO} is: BOD turn-on ($t_d \approx 0.3\,\mu$s), rise of the gate current to the level set by R_1, thyristor turn-on after t_{gd} (a few µs for large phase-control thyristors), collapse of v_{AK}. During $t_d + t_{gd}$ the anode voltage keeps rising at the rate imposed by the circuit, so the peak voltage seen by the thyristor is

$$V_{pk} \approx V_{\mathrm{BO}} + \left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{\mathrm{circuit}} (t_d + t_{gd}), \tag{2}$$

which must stay below V_{DRM} of the thyristor. With the 100 V/µs limit of the DTT pre-filter [8] and $t_d + t_{gd} \approx 4\,\mu$s the overshoot is about 400 V.

3.2 Design rules

1. *Choice of V_{BO}.* The BOD must not fire during normal operation and must fire before the thyristor breaks over:

$$V_{op,max} + \Delta V_{margin} \le V_{\mathrm{BO}}^{min}(T_{\mathrm{j}}), \qquad V_{\mathrm{BO}}^{max}(T_{\mathrm{j}}) + \left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{\mathrm{circuit}} (t_d + t_{gd}) \le V_{\mathrm{DRM}}, \tag{3}$$

   where $V_{\mathrm{BO}}^{min,max}$ include the ±50 V tolerance and the temperature coefficient K_T over the operating range (+12.5 % at 150 °C with $K_T = 1 \times 10^{-3}$/K). In the DTT option-B configuration the shunt regulator level was 920 V and the backup BOD 1100 V [8]: the minimum breakover 1050 V at 25 °C leaves 130 V of margin; at −40 °C the margin shrinks to about 60 V.

2. *Series stacking.* Higher voltages use n elements in series (modules with 2–4 dice are standard): V_{BO} and V_T add ($V_T = 1.7\,n$ V for the IXBOD1 modules), the tolerance grows as $n \times 50$ V and the static voltage sharing is provided by parallel resistors R_p across each element, sized so that their current is large compared with the spread of the off-state currents. Series connection increases the $\mathrm{d}v/\mathrm{d}t$ capability (the junction capacitances are in series) [2].

3. *Current-limiting resistor R_1.* It sets the gate current and the BOD current: $i_G \approx (V_{\mathrm{BO}} - V_{GK})/R_1$ must be several times I_{GT} of the thyristor for a fast, safe turn-on but below the peak forward gate current I_{FGM} (10 A for the 5STP 26N6500 used in the static-switch bench [9]) and within the BOD surge and $\mathrm{d}i/\mathrm{d}t$ limits. Values of 0–200 Ω are recommended [2]; 100–150 Ω

give 7–11 A with an 1100 V BOD. R_1 also damps the loop formed by the gate capacitance C_2 and the lead inductances.

4. *Series diode D_1.* It must block the full reverse voltage applied to the thyristor (anode negative), since the BOD withstands only $V_{RB} \approx 10\,\mathrm{V}$, and it must be a fast, soft-recovery diode: at the end of each BOD pulse the reverse recovery of D_1 is what drives the BOD into reverse. Its avalanche capability must cover the recovery of the thyristor.

5. *Gate network.* $R_2 \parallel C_2$ (100–1000 Ω, 22–47 nF) keeps the gate at low impedance for noise and for the currents injected through the anode–gate capacitances at high $\mathrm{d}v/\mathrm{d}t$. It does not delay the firing: the gate–cathode junction clamps the gate voltage at about V_{GT}, so C_2 only has to be charged to a few volts, which a trigger current of several amperes does in tens of nanoseconds (Fig. 6).

6. *Energy and repetition.* The energy per event in the BOD is small: about $\frac{1}{2} V_{\mathrm{BO}}\, i\, t_d$ during the voltage collapse plus $V_T\, i_G\, t_{gd}$ in the on-state, i.e. 0.2–0.3 mJ at 10 A (Section 5), a temperature rise $K_P E$ of a fraction of a kelvin. The energy goes into R_1 instead, $E_{R1} \approx V_{\mathrm{BO}}\, i_G\, (t_d + t_{gd}) \approx 40\,\mathrm{mJ}$ per event at 10 A, which requires a pulse-rated resistor. The average dissipation is negligible at the repetition rates of a crowbar, but the junction temperature before the event (ambient, self-heating of the resistors nearby) shifts V_{BO} through K_T.

7. *Spurious triggering by $\mathrm{d}v/\mathrm{d}t$.* The BOD sees the anode $\mathrm{d}v/\mathrm{d}t$ through D_1 and the gate network; with the switching edges of an IGBT bridge (kV/µs) a stack close to its V_{BO} fires even if the voltage stays below V_{BO}. This is exactly what the DTT crowbar did at the first switch-on [8]. The remedies are the pre-filter (limit the $\mathrm{d}v/\mathrm{d}t$ at the crowbar node), a small R–C across the BOD branch, or a larger margin between the operating voltage and V_{BO}: the criterion of Section 4.2 shows that the trigger level scales as $0.9\, V_{\mathrm{BO}}\, (\mathrm{d}v/\mathrm{d}t)_{cr}/(\mathrm{d}v/\mathrm{d}t)$.

4 The Simscape model `dtt.crowbar.bod`

4.1 Structure and modes

The component (Fig. 4) has the electrical ports `p` (anode) and `n` (cathode), the physical-signal output `M` (mode indicator) and the optional thermal port `H`. Electrically it is a conduction current $i_{dev}(v, \text{mode})$ in parallel with the junction capacitance C_j:

$$i = i_{dev} + C_j \frac{\mathrm{d}v}{\mathrm{d}t}. \tag{4}$$

The capacitance makes v a state in every mode, which keeps the equation structure identical across modes, provides the $\mathrm{d}v/\mathrm{d}t$ needed by the $\mathrm{d}v/\mathrm{d}t$ trigger criterion and gives the device voltage a physical continuity at the mode transitions. The mode chart has four modes:

- `Blocking` ($M = 0$): forward blocking with off-state current and pre-breakover term, reverse breakdown beyond $-V_{RB}$; the blocking capability x_b recovers towards 1.
- `Triggered` ($M = 1$) and `TriggeredDV` ($M = 2$): turn-on after a voltage or a $\mathrm{d}v/\mathrm{d}t$ trigger; the device acts as a clamp at the voltage $v_{cl}(x_b)$, which collapses exponentially from $V_{\mathrm{BO}}(T_\mathrm{j})$ to V_{T0} while x_b decays from 1 to $x_{on} = 10^{-3}$ in the time t_d.
- `On` ($M = 3$): on-state $v = V_{\mathrm{T0}} + r_{\mathrm{T}}\, i_{dev}$, left when $i_{dev} < I_{\mathrm{H}}(T_\mathrm{j})$.

The blocking-capability state $x_b \in [0, 1]$ is the single internal state besides v: it drives the turn-on collapse (`Triggered` and `On`) and the recovery after turn-off (`Blocking`), and it defines the actual breakover level $V_{blk}(x_b)$ during recovery. No `delay()` function and no `initial` section are used, in line with the constraints found in the `+dtt/+crowbar` package [8]: the first declared mode, `Blocking`, is the initial one.

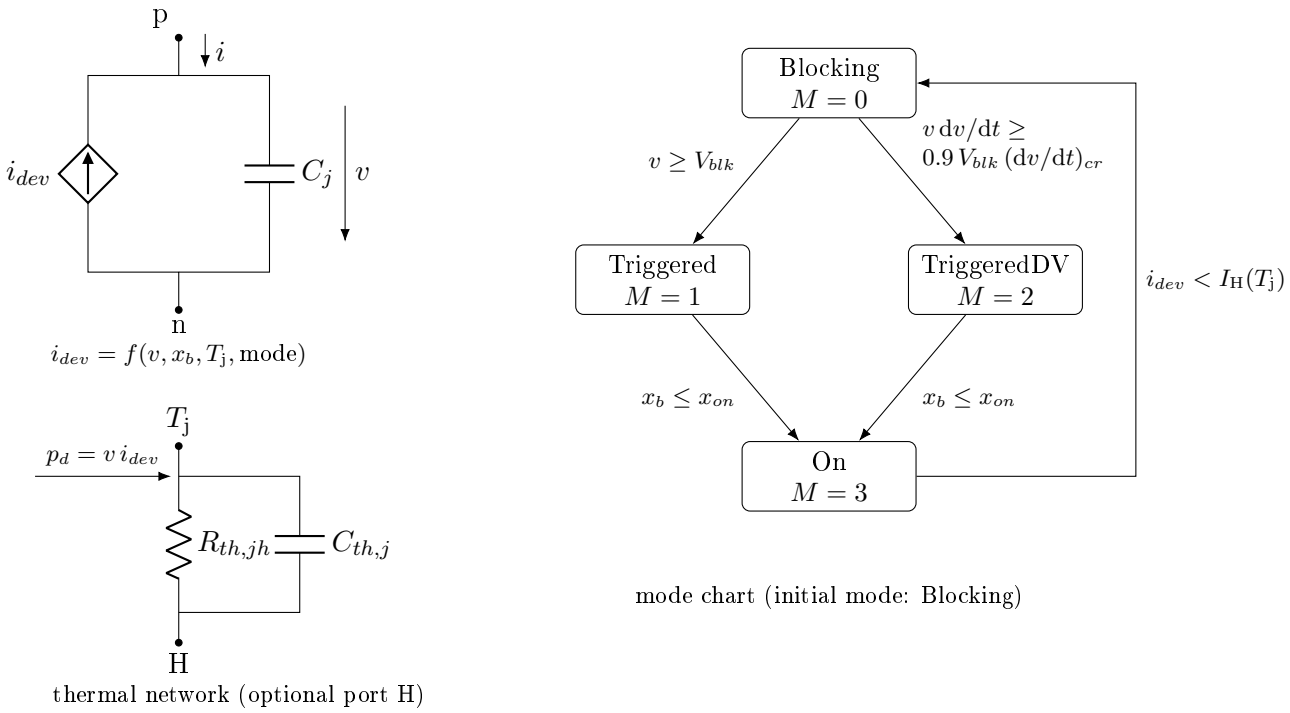


Figure 4: Structure of the model: conduction current i_{dev} in parallel with C_j; thermal network between the junction and the port H (only with `thermal_port = on`); mode chart with the transition predicates.

4.2 Equations

Temperature dependence. With $\Delta T = T_\mathrm{j} - 25\,^\circ\mathrm{C}$,

$$V_\mathrm{BO}(T_\mathrm{j}) = V_\mathrm{BO}\,(1 + K_T\,\Delta T), \tag{5}$$

$$I_\mathrm{H}(T_\mathrm{j}) = I_\mathrm{H}^{25}\,e^{-k_H \Delta T}, \qquad k_H = \frac{\ln\!\left(I_\mathrm{H}^{25}/I_\mathrm{H}^{125}\right)}{100\,\mathrm{K}}, \tag{6}$$

$$I_D(T_\mathrm{j}) = I_D^{25}\,e^{k_D \Delta T}, \qquad k_D = \frac{\ln\!\left(I_D^{125}/I_D^{25}\right)}{100\,\mathrm{K}}, \tag{7}$$

$$(\mathrm{d}v/\mathrm{d}t)_{cr}(T_\mathrm{j}) = (\mathrm{d}v/\mathrm{d}t)_{cr}^{25}\,e^{-k_{dv} \Delta T}, \qquad k_{dv} = \frac{\ln\!\left((\mathrm{d}v/\mathrm{d}t)_{cr}^{25}/(\mathrm{d}v/\mathrm{d}t)_{cr}^{150}\right)}{125\,\mathrm{K}}. \tag{8}$$

The exponential laws pass exactly through the two datasheet points of each quantity (Fig. 9); the off-state current doubles every 10 K, as expected for a leakage dominated by generation.

Blocking capability. The state x_b obeys

$$\frac{\mathrm{d}x_b}{\mathrm{d}t} = \begin{cases} (1 - x_b)/\tau_q, & \tau_q = t_q/\ln 100 \quad \text{(Blocking)} \\ -x_b/\tau_d, & \tau_d = t_d/\ln(1/x_{on}) \quad \text{(Triggered, TriggeredDV, On)} \end{cases} \tag{9}$$

so that in `Blocking` it recovers to 99 % in t_q, and in the turn-on modes it decays from 1 to $x_{on} = 10^{-3}$ in exactly t_d. The actual breakover level and the turn-on clamp are

$$V_{blk} = V_{rq} + x_b\left(V_\mathrm{BO}(T_\mathrm{j}) - V_{rq}\right), \qquad v_{cl} = V_\mathrm{T0} + x_b\left(V_\mathrm{BO}(T_\mathrm{j}) - V_\mathrm{T0}\right). \tag{10}$$

Immediately after a turn-off ($x_b \approx 0$) the device therefore blocks only V_{rq} (assumed 50 V); after t_q it blocks $0.99\,V_\mathrm{BO}$ and after $2\,t_q$ essentially V_BO. The floor V_{rq} is needed to avoid an immediate re-trigger at the on-state voltage left on C_j after turn-off.

Blocking mode.

$$i_{dev} = \begin{cases} \dfrac{I_D(T_\mathrm{j})}{0.8\,V_\mathrm{BO}(T_\mathrm{j})}\,v + I_\mathrm{BO}\left(\dfrac{v}{V_\mathrm{BO}(T_\mathrm{j})}\right)^m, & v \geq 0 \\ \dfrac{I_D(T_\mathrm{j})}{0.8\,V_\mathrm{BO}(T_\mathrm{j})}\,v, & -V_{RB} \leq v < 0 \\ \dfrac{I_D(T_\mathrm{j})}{0.8\,V_\mathrm{BO}(T_\mathrm{j})}\,v + \dfrac{v + V_{RB}}{R_{RB}}, & v < -V_{RB} \end{cases} \tag{11}$$

The linear term reproduces the datasheet off-state current at $0.8\,V_\mathrm{BO}$; the power term with $m = 20$ mimics the avalanche multiplication and reaches I_BO at $V_\mathrm{BO}(T_\mathrm{j})$ (Fig. 5). The breakover event is

$$v \geq V_{blk}(x_b) \quad \Rightarrow \quad \text{Triggered.} \tag{12}$$

$\mathrm{d}v/\mathrm{d}t$ **criterion.** The displacement current $C_j\,\mathrm{d}v/\mathrm{d}t$ multiplied by a voltage-dependent gain is the physical trigger; the model uses the simplest voltage-dependent form, a product, calibrated at the datasheet point:

$$v > 0 \;\wedge\; v\,\frac{\mathrm{d}v}{\mathrm{d}t} \geq 0.9\,V_{blk}(x_b)\,(\mathrm{d}v/\mathrm{d}t)_{cr}(T_\mathrm{j}) \quad \Rightarrow \quad \text{TriggeredDV.} \tag{13}$$

At $v = 0.9\,V_\mathrm{BO}$ the device fires exactly at $(\mathrm{d}v/\mathrm{d}t)_{cr}$; at half the voltage it needs twice the rate; at $v = V_\mathrm{BO}$ the voltage trigger prevails anyway. With a ramp of slope S the $\mathrm{d}v/\mathrm{d}t$ trigger occurs at $v = 0.9\,V_\mathrm{BO}(\mathrm{d}v/\mathrm{d}t)_{cr}/S$ if this is below V_BO, i.e. the ramp must exceed $0.9\,(\mathrm{d}v/\mathrm{d}t)_{cr}$ to trigger the device before V_BO.

Turn-on modes. The device clamps at $v_{cl}(x_b)$ through the dynamic resistance r_{av}:

$$i_{dev} = \frac{I_D(T_\mathrm{j})}{0.8\,V_\mathrm{BO}(T_\mathrm{j})}\,v + \frac{\max(0,\, v - v_{cl})}{r_{av}}. \tag{14}$$

With $x_b = 1$ at the trigger instant $v_{cl} = V_\mathrm{BO}(T_\mathrm{j})$, so the current at the beginning of the collapse is fixed by the external circuit; v_{cl} then falls exponentially with τ_d and the mode ends when $x_b \leq x_{on}$, i.e. when v_{cl} has reached $V_\mathrm{T0} + 10^{-3}(V_\mathrm{BO} - V_\mathrm{T0}) \approx 2\,\mathrm{V}$. The residual step to the on-state characteristic is thus about 1 V, which keeps the internal discharge of C_j (time constant $r_\mathrm{T}C_j$, a few picoseconds) negligible.

On-state and turn-off.

$$i_{dev} = \frac{v - V_\mathrm{T0}}{r_\mathrm{T}}, \qquad i_{dev} < I_\mathrm{H}(T_\mathrm{j}) \;\Rightarrow\; \text{Blocking.} \tag{15}$$

The holding voltage V_H is not represented: the on-state line is used down to I_H. A reverse current forces $i_{dev} < 0 < I_\mathrm{H}$ and the device returns to `Blocking`, where the reverse branch of Eq. (11) applies.

Thermal port. With `thermal_port = on` the junction temperature is a state:

$$C_{th,j}\,\frac{\mathrm{d}T_\mathrm{j}}{\mathrm{d}t} = v\,i_{dev} - \frac{T_\mathrm{j} - T_H}{R_{th,jh}}, \qquad C_{th,j} = \frac{1}{K_P} = 1.43\,\mathrm{mJ/K}, \tag{16}$$

where T_H is the temperature of the node connected to H (ambient, with $R_{th,jh} = R_{th(j\text{-}a)} = 60\,\mathrm{K/W}$ as default, or the tab if a finer network is used). The dissipation is $v\,i_{dev}$, without the reactive current of C_j. With the port off, T_j is the fixed parameter `T_j`.

Table 3: Parameters of `dtt.crowbar.bod` (defaults: IXBOD2-11)

Name	Symbol	Default	Source
`V_BO`	V_{BO} (25 °C)	1100 V	datasheet
`I_BO`	I_{BO}	20 mA	datasheet
`m_av`	m, exponent of the pre-breakover term	20	assumed (shape only)
`I_D25`, `I_D125`	I_D at $0.8V_{\mathrm{BO}}$, 25 / 125 °C	0.2 / 200 µA	datasheet
`V_T0`, `r_T`	V_{T0}, r_{T}	0.9 V, 80 mΩ	datasheet
`I_H25`, `I_H125`	I_{H} at 25 / 125 °C	40 / 20 mA	datasheet
`K_T`	K_T	1×10^{-3}/K	datasheet
`dvdt_cr25`, `dvdt_cr150`	$(\mathrm{d}v/\mathrm{d}t)_{cr}$ at 25 / 150 °C	3000 / 1000 V/µs	datasheet
`V_RB`	V_{RB}	10 V	IXYS application note
`R_RB`	R_{RB}	1 Ω	assumed
`C_j`	C_j	20 pF	assumed
`t_d`	t_d, turn-on time	0.3 µs	assumed (patents)
`r_av`	r_{av}	1 Ω	assumed
`t_q`	t_q	200 µs	datasheet
`V_rq`	V_{rq}, blocking capability after turn-off	50 V	assumed
`T_j`	T_{j} (thermal port off)	25 °C	–
`thermal_port`	–	off	–
`R_thjh`	$R_{th,jh}$	60 K/W	datasheet ($R_{th(j\text{-}a)}$)
`C_thj`	$C_{th,j} = 1/K_P$	1.43 mJ/K	datasheet (K_P)

4.3 Parameters

Table 3 lists the parameters with the default values (IXBOD2-11) and their source. The private parameters k_H, k_D, k_{dv}, τ_d, τ_q are derived in the `parameters(Access=private)` section.

4.4 Implementation notes

- The equation structure is the same in all modes (one conduction equation, one state equation for x_b, one output equation), as required by Simscape mode charts; v is a state in every mode through C_j, therefore `C_j` must be positive (asserted).
- v is declared without initialization priority, so that the solver can start from a consistent voltage when the BOD is connected across a charged node; x_b starts at 1 (high priority) and T_{j} at 25 °C (high priority, thermal port on): both can be overridden in the *Variables* tab of the block.
- The output `M` is an algebraic physical signal (0/1/2/3) suitable for logging the trigger cause; `i_dev`, `xb`, `V_blk`, `V_BOT`, `I_HT` and the mode chart are observable in the Simscape Results Explorer.
- The turn-on collapse has the time constant $\tau_d = t_d/6.9 \approx 43$ ns and the clamp follows v_{cl} with $r_{av}C_j = 20$ ps: a stiff but smooth transient that the variable-step DAE solvers (`daessc`, `ode23t`, `ode15s`) handle without difficulty. The relaxation oscillation that a real BOD shows when fed from a current source between I_{BO} and I_{H} is reproduced by the model (period $\approx C_j V_{rq}/i$); with a voltage source and a load line it does not occur.
- The private derived parameters use `log()` of dimensionless ratios; the public junction temperature `T_j` is declared in °C, the private reference temperatures in kelvin, so that the temperature differences are unambiguous for the unit system.
- If the conditional thermal section or the enumeration `simscape.enum.onoff` is not accepted by the installed release, the fallback is to remove the `if`/`else` block and keep the `else` branch (`Tj == T_j`).

4.5 Limitations

The model has no charge dynamics: the on-state, the turn-on collapse and the recovery are represented by the surrogate state x_b rather than by stored charge, so the reverse recovery of the BOD (relevant only through D_1), the current-dependent turn-on and the dependence of t_q on the previous current are not represented. The $\mathrm{d}v/\mathrm{d}t$ criterion is a calibrated approximation valid near the rated point; the $\mathrm{d}i/\mathrm{d}t$ limit, the surge I^2t and the holding voltage V_H are not enforced (they must be checked from the logged waveforms). The pre-breakover exponent m, C_j, t_d, r_{av}, V_{rq} and R_{RB} are assumed values chosen for a plausible shape and numerical robustness, not fitted to measurements; the OCEM test report on the DTT BOD board [8] could be used to refine t_d and the $\mathrm{d}v/\mathrm{d}t$ behaviour.

5 Verification with the Python replica

The script `bod_sim.py` implements the equations of Section 4 (same parameter names, same mode logic with event detection) and integrates them together with small external circuits using a stiff solver (Radau). It is a check of the model logic and of the choice of the assumed parameters, not a substitute for the Simscape compilation.

Static characteristic (Fig. 5). A curve-tracer sweep: source 0–1300 V and back through $R_L = 1\,\mathrm{k\Omega}$. On the rising leg the conduction current follows the off-state line (0.2 µA at 880 V) and the avalanche term up to $(V_\mathrm{BO}, I_\mathrm{BO})$; the source must reach $1120\,\mathrm{V} = V_\mathrm{BO} + I_\mathrm{BO}R_L$ for the breakover to occur, then the operating point jumps along the load line to the on-state at 1.1 A. On the falling leg the device stays on down to $I_\mathrm{H} = 40\,\mathrm{mA}$ (source at 41 V) and jumps back to the blocking branch.

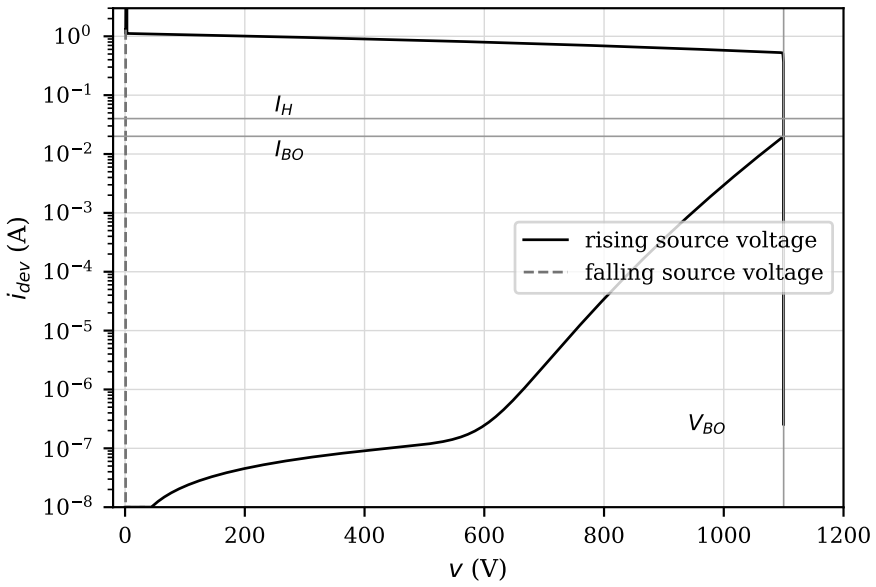


Figure 5: Static characteristic of the model (conduction current, logarithmic scale) obtained with a triangular source sweep through 1 kΩ; the vertical segments are the jumps along the load line at breakover and at turn-off.

Protection of a thyristor (Fig. 6). Circuit of Fig. 3 with a source ramping at 200 V/µs to 1600 V through $R_s = 2\,\Omega$, $R_1 = 150\,\Omega$, $R_2 = 220\,\Omega$, $C_2 = 22\,\mathrm{nF}$, ideal D_1, and a functional thyristor with $V_{GT} = 2.5\,\mathrm{V}$, $I_{GT} = 0.3\,\mathrm{A}$, $t_{gd} = 3\,\mu\mathrm{s}$. The BOD triggers at $t = 5.52\,\mu\mathrm{s}$ ($v_s = 1104\,\mathrm{V}$, i.e. V_BO plus the drops), the gate current exceeds I_{GT} 25 ns later (the collapse of v_{cl} starts immediately), the BOD is fully on after $t_d = 0.3\,\mu\mathrm{s}$ and carries about 10 A (set by R_1) until the thyristor turns on at 8.55 µs; the anode voltage, which had continued to rise to 1600 V during $t_d + t_{gd}$, collapses and the

BOD turns off within the same step. The energy dissipated in the BOD is 0.25 mJ (0.2 mJ during the 0.3 µs collapse, 42 µJ in the on-state), against 38 mJ in R_1.

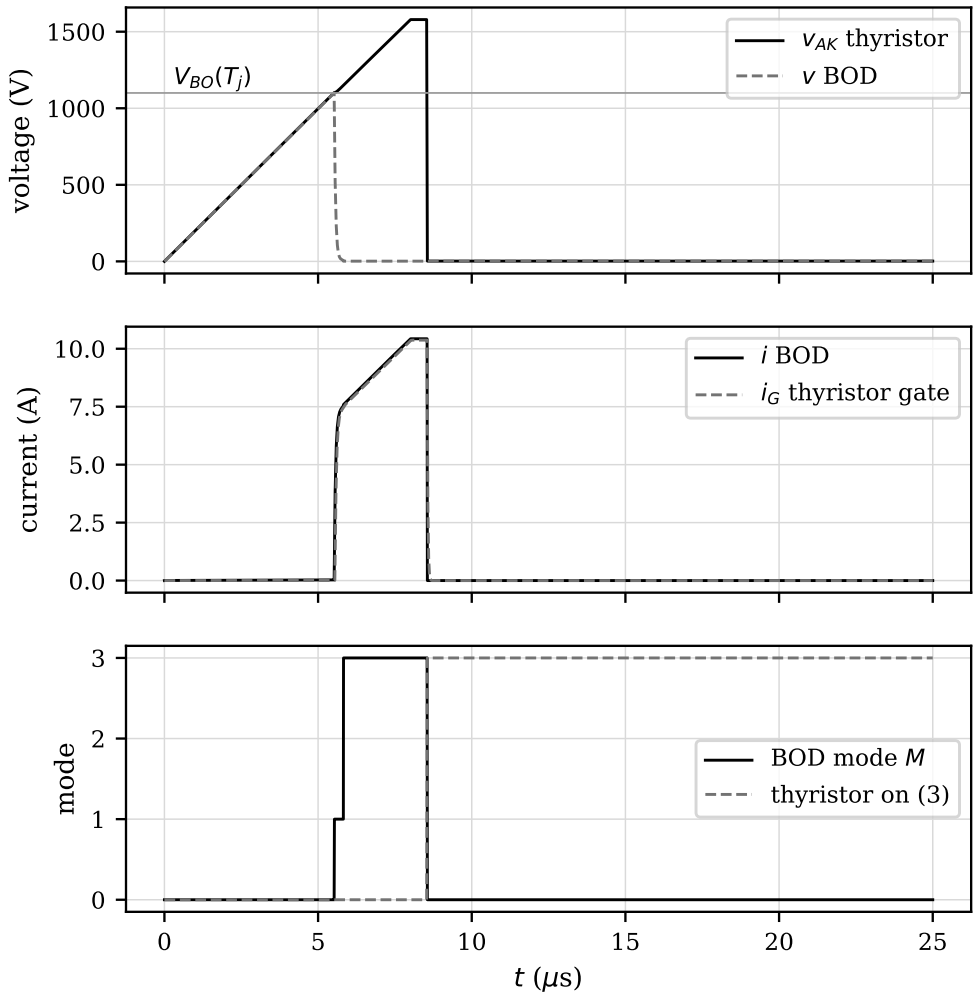


Figure 6: BOD-triggered turn-on of a thyristor (Python replica): anode and BOD voltages, BOD and gate currents, BOD mode and thyristor state.

$\mathrm{d}v/\mathrm{d}t$ triggering (Fig. 7). Voltage ramps through 10 Ω up to $0.95\,V_{\mathrm{BO}}$ (1045 V, below V_{BO} at both temperatures): 1500 V/µs at 25 °C does not trigger; 4000 V/µs triggers at $743\,\mathrm{V} = 0.9 \cdot 1100 \cdot 3000/4000$; at 150 °C, with $(\mathrm{d}v/\mathrm{d}t)_{cr} = 1000\,\mathrm{V/\mu s}$ and $V_{\mathrm{BO}} = 1238\,\mathrm{V}$, 1500 V/µs triggers at 742 V. The voltage then collapses in 0.3 µs.

Recovery after turn-off (Fig. 8). The BOD conducts for 20 µs (source 1300 V through 20 Ω), the source is removed and a ramp of 200 V/µs is re-applied 50 µs or 500 µs later. In the first case $x_b = 0.71$ and the device re-triggers at $V_{blk} = 797\,\mathrm{V}$; in the second $x_b \approx 1$ and it re-triggers at 1100 V.

Temperature dependence (Fig. 9). V_{BO}, I_{H}, I_D and $(\mathrm{d}v/\mathrm{d}t)_{cr}$ from Eqs. (5)–(8) between −40 °C and 150 °C, with the datasheet points.

6 Use in the crowbar models

In the `+dtt/+crowbar` package the BOD is connected between the anode and the gate node of `thyristor_fn` (or of `thyristor_rr` in the detailed version) through a Diode block (or `diode_fn`)

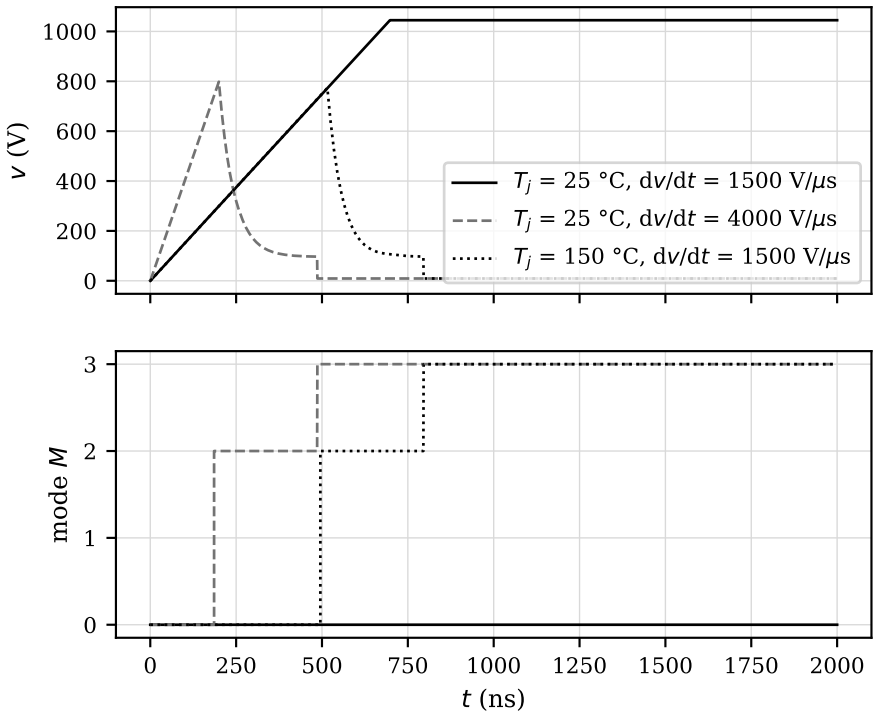


Figure 7: dv/dt triggering below V_{BO}: ramps of 1500 and 4000 V/µs at 25 °C and 1500 V/µs at 150 °C, all limited to 0.95 V_{BO}.

and a resistor R_1, as in Fig. 3; the gate node carries the $R_2 \parallel C_2$ network and the firing-unit output. For a self-fired crowbar without firing unit the same connection is the only gate source. The output `M` can be logged or fed to a Scope through a PS-Simulink converter to see whether a firing was caused by voltage ($M = 1$) or by dv/dt ($M = 2$), which is the diagnostic needed for the spurious firings observed on the machine.

For a stack of n elements one block is used with `V_BO`, `V_T0`, `V_RB`, `r_T` and `r_av` multiplied by n, `C_j` divided by n, and the parallel sharing resistors added externally. With the thermal port on, a Thermal Mass and a Temperature Source (ambient) connected to H give the junction temperature and, through K_T, the drift of the firing level.

Recommended solver settings are those already used for the package: variable-step `daessc` or `ode23t`, relative tolerance 10^{-4} or tighter, and a maximum step small enough to resolve the 0.3 µs collapse when its waveform is of interest.

A Listing of `bod.ssc`

```
component bod
% Break-over diode (BOD)
% Two-terminal p-n-p-n break-over diode used as a passive overvoltage
% trigger for thyristors (anode-to-gate connection).
%
% Functional model with switching dynamics:
%  - forward blocking: temperature-dependent off-state current plus a
%    steep pre-breakover (avalanche-multiplication) term reaching I_BO
%    at the break-over voltage V_BO(T) = V_BO*(1 + K_T*(T - 25 degC));
%  - break-over event at v >= V_BO(T) or by dv/dt (displacement current
%    criterion calibrated at the datasheet point 0.9*V_BO, dv/dt_cr(T));
%  - turn-on: the voltage collapses from V_BO to the on-state along an
%    exponential with total duration t_d (blocking-capability state xb);
%  - on-state: v = V_T0 + r_T*i, turn-off when i < I_H(T);
%  - recovery: after turn-off the blocking capability rebuilds from V_rq
```

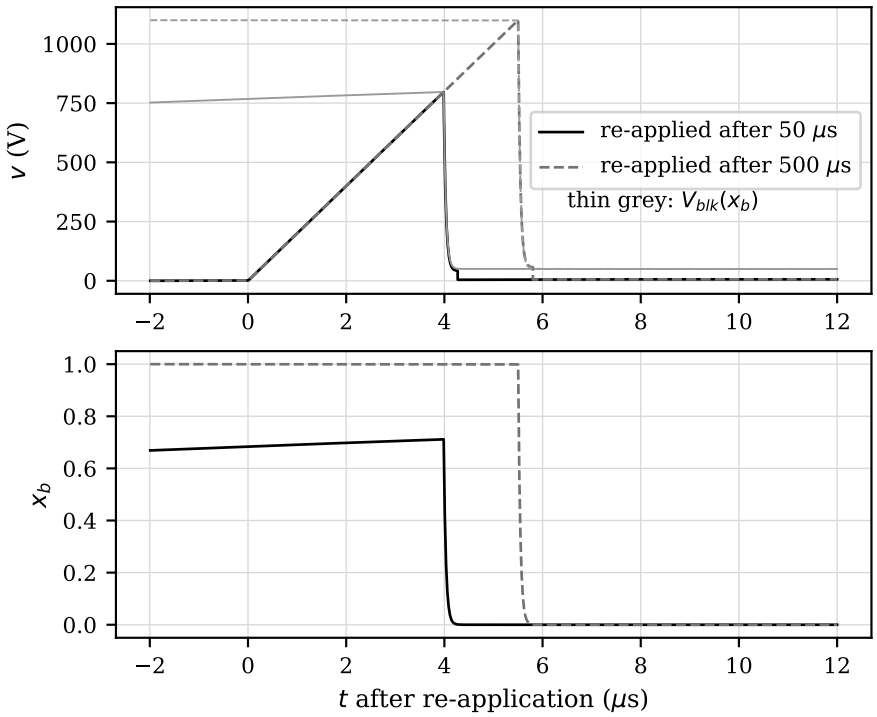


Figure 8: Re-applied voltage after turn-off: reduced breakover level during the recovery (thin grey line, $V_{blk}(x_b)$) and full V_{BO} after more than $2\,t_q$.

```
%    to V_BO(T) with the time constant t_q/ln(100) (tq surrogate);
%    a forward voltage re-applied earlier re-triggers the device;
%  - reverse: the BOD blocks only V_RB (about 10 V), beyond that it breaks
%    down with the resistance R_RB (a series diode is required in the
%    application circuit);
%  - junction capacitance C_j in parallel (also provides dv/dt);
%  - optional thermal port H: junction thermal mass C_thj = 1/K_P and
%    junction-to-port resistance R_thjh, dissipation p = v*i_dev.
%
% Output M: 0 blocking, 1 break-over (voltage) triggered, 2 dv/dt
% triggered, 3 on-state.
%
% Default parameters: Littelfuse/IXYS IXBOD2-11 (V_BO = 1100 V +/- 50 V).
% Values marked (assumed) are not given in the datasheet.
%
% Package dtt.crowbar - functional models for the DTT DIV crowbar.
% Rev. 00, 2026-09-22.

nodes
    p = foundation.electrical.electrical; % +:left
    n = foundation.electrical.electrical; % -:right
end

outputs
    M = {0, '1'}; % M:right
end

parameters
    V_BO      = {1100, 'V'};     % Break-over voltage V_BO at 25 degC
    I_BO      = {20,   'mA'};    % Break-over current I_BO
    m_av      = {20,   '1'};     % Exponent of the pre-breakover current
    I_D25     = {0.2,  'uA'};    % Off-state current at 0.8*V_BO, 25 degC
    I_D125    = {200,  'uA'};    % Off-state current at 0.8*V_BO, 125 degC
    V_T0      = {0.9,  'V'};     % On-state threshold voltage V_T0
```

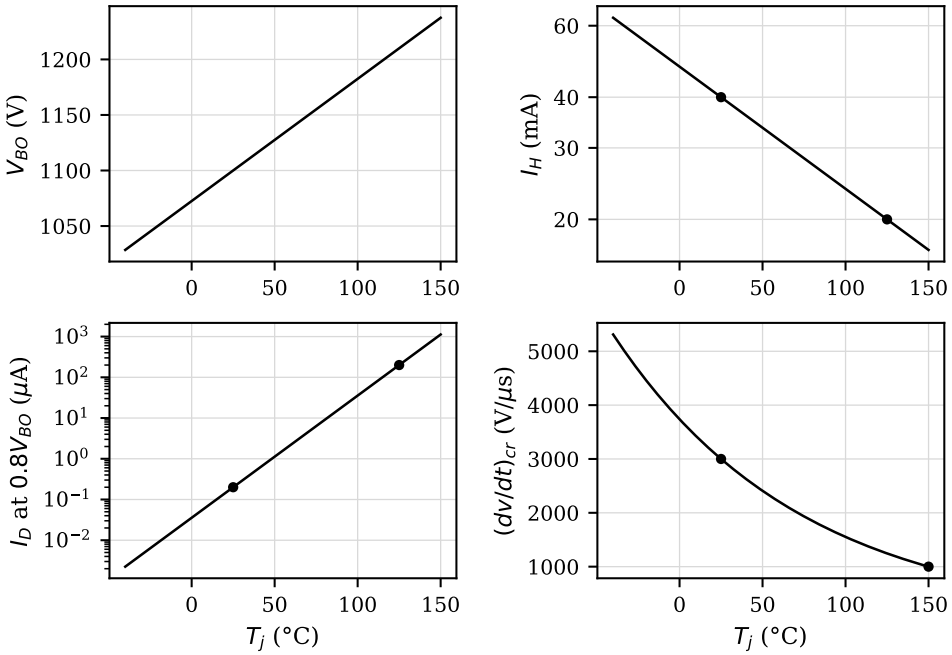


Figure 9: Temperature dependence used in the model; dots are the datasheet points.

```
    r_T       = {80,   'mOhm'};  % On-state slope resistance r_T
    I_H25     = {40,   'mA'};    % Holding current I_H at 25 degC
    I_H125    = {20,   'mA'};    % Holding current I_H at 125 degC
    K_T       = {1e-3, '1/K'};   % Temperature coefficient of V_BO, K_T
    dvdt_cr25 = {3000, 'V/us'};  % Critical dv/dt at 0.9*V_BO, 25 degC
    dvdt_cr150= {1000, 'V/us'};  % Critical dv/dt at 0.9*V_BO, 150 degC
    V_RB      = {10,   'V'};     % Reverse blocking (breakdown) voltage
    R_RB      = {1,    'Ohm'};   % Reverse breakdown resistance (assumed)
    C_j       = {20,   'pF'};    % Junction capacitance (assumed)
    t_d       = {0.3,  'us'};    % Turn-on time, V_BO to on-state (assumed)
    r_av      = {1,    'Ohm'};   % Dynamic resistance during turn-on (assumed)
    t_q       = {200,  'us'};    % Turn-off (recovery) time t_q
    V_rq      = {50,   'V'};     % Blocking capability right after turn-off (assumed)
    T_j       = {25,   'degC'};  % Junction temperature (thermal port off)
    thermal_port = simscape.enum.onoff.off; % Thermal port
    R_thjh    = {60,   'K/W'};   % Thermal resistance junction to port H
    C_thj     = {1.43, 'mJ/K'};  % Junction thermal mass (1/K_P)
end

parameters (Access = private)
    T_ref  = {298.15, 'K'};  % Reference temperature (25 degC)
    T_hot  = {398.15, 'K'};  % Temperature of the I_D125 / I_H125 data (125 degC)
    T_hot2 = {423.15, 'K'};  % Temperature of the dvdt_cr150 data (150 degC)
    x_on   = {1e-3, '1'};    % Blocking-capability level at the end of turn-on
    k_H    = log(I_H25/I_H125)/(T_hot - T_ref);        % I_H decay rate
    k_D    = log(I_D125/I_D25)/(T_hot - T_ref);        % I_D growth rate
    k_dv   = log(dvdt_cr25/dvdt_cr150)/(T_hot2 - T_ref); % dv/dt_cr decay rate
    tau_d  = t_d/log(1/x_on);                          % Turn-on time constant
    tau_q  = t_q/log(100);                             % Recovery time constant (99 % at t_q)
end

variables
    v       = {0, 'V'};    % Voltage
    i       = {0, 'A'};    % Current
    i_dev   = {0, 'A'};    % Conduction current (without C_j)
    dvdt    = {0, 'V/s'};  % dv/dt
    xb      = {value = {1, '1'}, priority = priority.high}; % Blocking capability (0..1)
    V_BOT   = {1100, 'V'}; % Break-over voltage at T_j
```

```
    V_blk   = {1100, 'V'}; % Actual blocking capability
    v_cl    = {1100, 'V'}; % Clamp voltage during turn-on
    I_HT    = {40, 'mA'};  % Holding current at T_j
    I_DT    = {0.2, 'uA'}; % Off-state current at T_j
    dvdt_crT= {3000, 'V/us'}; % Critical dv/dt at T_j
    p_d     = {0, 'W'};    % Dissipated power
end

branches
    i : p.i -> n.i;
end

if thermal_port == simscape.enum.onoff.on
    nodes
        H = foundation.thermal.thermal; % H:bottom
    end
    variables
        Tj = {value = {25, 'degC'}, priority = priority.high}; % Junction temperature
        Q  = {0, 'W'}; % Heat flow from port H into the junction
    end
    branches
        Q : H.Q -> *;
    end
    equations
        Q == -(Tj - H.T)/R_thjh;
        C_thj*der(Tj) == p_d + Q;
    end
else
    variables
        Tj = {25, 'degC'}; % Junction temperature
    end
    equations
        Tj == T_j;
    end
end

equations
    assert(C_j > 0, 'The junction capacitance C_j must be positive')
    assert(t_d > 0, 'The turn-on time t_d must be positive')
    assert(t_q > 0, 'The turn-off time t_q must be positive')
    assert(V_rq > V_T0 && V_rq < V_BO, 'V_rq must lie between V_T0 and V_BO')
    assert(V_RB > 0, 'The reverse blocking voltage V_RB must be positive')

    v == p.v - n.v;
    dvdt == der(v);
    i == i_dev + C_j*dvdt;

    % Temperature dependence
    V_BOT    == V_BO*(1 + K_T*(Tj - T_ref));
    I_HT     == I_H25*exp(-k_H*(Tj - T_ref));
    I_DT     == I_D25*exp(k_D*(Tj - T_ref));
    dvdt_crT == dvdt_cr25*exp(-k_dv*(Tj - T_ref));

    % Blocking capability and turn-on clamp
    V_blk == V_rq + xb*(V_BOT - V_rq);
    v_cl  == V_T0 + xb*(V_BOT - V_T0);

    p_d == v*i_dev;
end

modecharts (ExternalAccess = observe)
    bod_mc = modechart
        modes
            mode Blocking
                equations
```

```
                    if v >= 0
                        i_dev == I_DT*v/(0.8*V_BOT) + I_BO*(v/V_BOT)^m_av;
                    elseif v >= -V_RB
                        i_dev == I_DT*v/(0.8*V_BOT);
                    else
                        i_dev == I_DT*v/(0.8*V_BOT) + (v + V_RB)/R_RB;
                    end
                    der(xb) == (1 - xb)/tau_q;
                    M == 0;
                end
            end
            mode Triggered
                equations
                    if v > v_cl
                        i_dev == I_DT*v/(0.8*V_BOT) + (v - v_cl)/r_av;
                    else
                        i_dev == I_DT*v/(0.8*V_BOT);
                    end
                    der(xb) == -xb/tau_d;
                    M == 1;
                end
            end
            mode TriggeredDV
                equations
                    if v > v_cl
                        i_dev == I_DT*v/(0.8*V_BOT) + (v - v_cl)/r_av;
                    else
                        i_dev == I_DT*v/(0.8*V_BOT);
                    end
                    der(xb) == -xb/tau_d;
                    M == 2;
                end
            end
            mode On
                equations
                    i_dev == (v - V_T0)/r_T;
                    der(xb) == -xb/tau_d;
                    M == 3;
                end
            end
        end
        transitions
            Blocking    -> Triggered   : v >= V_blk;
            Blocking    -> TriggeredDV : v > 0 && v*dvdt >= 0.9*V_blk*dvdt_crT;
            Triggered   -> On          : xb <= x_on;
            TriggeredDV -> On          : xb <= x_on;
            On          -> Blocking    : i_dev < I_HT;
        end
    end
end

end
```

References

[1] H. Lawatsch and J. Vitins, “Protection of thyristors against overvoltage with breakover diodes,” *IEEE Trans. Ind. Appl.*, vol. 24, no. 3, pp. 444–448, May/June 1988.

[2] IXYS Corporation, “Breakover diodes – applications,” application section of the IXBOD1 data book, 2000.

[3] Littelfuse Inc., “IXBOD2 breakover diode Gen2 (IXBOD2-07 . . . IXBOD2-12),” datasheet, 2026.

[4] Littelfuse Inc. / IXYS, "IXBOD1 breakover diode Gen1 (IXBOD1-06 ... IXBOD1-10)," datasheet.

[5] IXYS Corporation, "Packaged overvoltage protection circuit for triggering thyristors," U.S. Patent Application 2014/0332842 A1, Nov. 2014.

[6] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York: Springer, 2008, ch. 8 (thyristors).

[7] J. Lutz, H. Schlangenotto, U. Scheuermann, and R. De Doncker, *Semiconductor Power Devices: Physics, Characteristics, Reliability*. Berlin: Springer, 2011, ch. 8.

[8] D. Bagnara, "Crowbar for the DTT DIV in-vessel coil power supplies," technical note `crowbar_dtt_div.tex`, Rev. 03, and "Option B configuration," `optionB_configuration.tex`, Rev. 00, Secom, Sept. 2026.

[9] D. Bagnara, "Crowbar thyristor with passive self-firing network," technical note `crowbar_thyristor_selftrigger.tex`, Rev. 02, Sept. 2026.

[10] The MathWorks, Inc., *Simscape Language Guide*, sections "Mode Charts" and "Conditional Declarations," 2026.